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Java SE 27 & JDK 27

Class AbstractQueue<E>

java.lang.Object
  java.util.AbstractCollection<E>
    java.util.AbstractQueue<E>

Type Parameters:

E - the type of elements held in this queue

All Implemented Interfaces:

Iterable<E>, Collection<E>, Queue<E>

Direct Known Subclasses:

ArrayBlockingQueue, ConcurrentLinkedQueue, DelayQueue, LinkedBlockingDeque, LinkedBlockingQueue, LinkedTransferQueue, PriorityBlockingQueue, PriorityQueue, SynchronousQueue

---

```
public abstract class AbstractQueue<E>
extends AbstractCollection<E>
implements Queue<E>
```

This class provides skeletal implementations of some Queue operations. The implementations in this class are appropriate when the base implementation does *not* allow null elements. Methods add, remove, and element are based on offer, poll, and peek, respectively, but throw exceptions instead of indicating failure via false or null returns.

A Queue implementation that extends this class must minimally define a method Queue.offer(E) which does not permit insertion of null elements, along with methods Queue.peek(), Queue.poll(), Collection.size(), and Collection.iterator(). Typically, additional methods will be overridden as well. If these requirements cannot be met, consider instead subclassing AbstractCollection.

This class is a member of the Java Collections Framework.

Since:

1.5

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**AbstractQueue**()	Constructor for use by subclasses.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
boolean	**add**(E e)	Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions, returning `true` upon success and throwing an `IllegalStateException` if no space is currently available.
boolean	**addAll**(Collection<? extends E> c)	Adds all of the elements in the specified collection to this queue.
void	**clear**()	Removes all of the elements from this queue.
E	**element**()	Retrieves, but does not remove, the head of this queue.
E	**remove**()	Retrieves and removes the head of this queue.

Methods declared in class AbstractCollection

contains, containsAll, isEmpty, iterator, remove, removeAll, retainAll, toArray, toArray, toString

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Methods declared in interface Collection

`equals`, `hashCode`, `parallelStream`, `removeIf`, `size`, `spliterator`, `stream`, `toArray`

Methods declared in interface Iterable

`forEach`

Methods declared in interface Queue

`offer`, `peek`, `poll`

Constructor Details

AbstractQueue

```
protected AbstractQueue()
```

Constructor for use by subclasses.

Method Details

add

```
public boolean add(E e)
```

Inserts the specified element into this queue if it is possible to do so immediately without violating capacity restrictions, returning `true` upon success and throwing an `IllegalStateException` if no space is currently available.

This implementation returns `true` if `offer` succeeds, else throws an `IllegalStateException`.

Specified by:

`add` in interface `Collection<E>`

Specified by:

`add` in interface `Queue<E>`

Overrides:

`add` in class `AbstractCollection<E>`

Parameters:

e - the element to add

Returns:

`true` (as specified by `Collection.add(E)`)

Throws:

`IllegalStateException` - if the element cannot be added at this time due to capacity restrictions

`ClassCastException` - if the class of the specified element prevents it from being added to this queue

`NullPointerException` - if the specified element is null and this queue does not permit null elements

`IllegalArgumentException` - if some property of this element prevents it from being added to this queue

remove

`public E remove()`

Retrieves and removes the head of this queue. This method differs from `poll` only in that it throws an exception if this queue is empty.

This implementation returns the result of `poll` unless the queue is empty.

Specified by:

`remove` in interface `Queue<E>`

Returns:

the head of this queue

Throws:

`NoSuchElementException` - if this queue is empty

element

`public E element()`

Retrieves, but does not remove, the head of this queue. This method differs from `peek` only in that it throws an exception if this queue is empty.

This implementation returns the result of `peek` unless the queue is empty.

Specified by:

`element` in interface `Queue<E>`

Returns:

the head of this queue

Throws:

`NoSuchElementException` - if this queue is empty

clear

```
public void clear()
```

Removes all of the elements from this queue. The queue will be empty after this call returns.

This implementation repeatedly invokes `poll` until it returns `null`.

Specified by:

`clear` in interface `Collection<E>`

Overrides:

`clear` in class `AbstractCollection<E>`

addAll

```
public boolean addAll(Collection<? extends E> c)
```

Adds all of the elements in the specified collection to this queue. Attempts to addAll of a queue to itself result in `IllegalArgumentException`. Further, the behavior of this operation is undefined if the specified collection is modified while the operation is in progress.

This implementation iterates over the specified collection, and adds each element returned by the iterator to this queue, in turn. A runtime exception encountered while trying to add an element (including, in particular, a `null` element) may result in only some of the elements having been successfully added when the associated exception is thrown.

Specified by:

`addAll` in interface `Collection<E>`

Overrides:

`addAll` in class `AbstractCollection<E>`

Parameters:

`c` - collection containing elements to be added to this queue

Returns:

`true` if this queue changed as a result of the call

Throws:

`ClassCastException` - if the class of an element of the specified collection prevents it from being added to this queue

`NullPointerException` - if the specified collection contains a null element and this queue does not permit null elements, or if the specified collection is null

`IllegalArgumentException` - if some property of an element of the specified collection prevents it from being added to this queue, or if the specified collection is this queue

`IllegalStateException` - if not all the elements can be added at this time due to insertion restrictions

See Also:

`add(Object)`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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